Siddhant Solanki

Ph.D. student in Physics | Stanford University

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Education

Stanford University June 2026 – Present
Ph.D. in Physics, in progress

University of Maryland, College Park June 2024 – June 2026
Ph.D. studies in Astronomy; transferred to Stanford

University of Maryland, College Park Aug. 2022 – June 2024
M.S. in Astronomy

University of California, Santa Barbara Sept. 2017 – Dec. 2020
B.S. in Physics

Research interests

Waves and instabilities; wave–particle interactions; relativistic plasmas; magnetars and neutron stars; fast radio bursts; black hole magnetospheres.

Methods and technical skills

Programming and computing: C, C++, Python; OpenMP, MPI, Kokkos; high-performance computing; Git and GitHub.
Modeling and analysis: Force-free, particle-in-cell, and magnetohydrodynamic simulations; analytical modeling of complex systems; general relativistic ray tracing; ParaView and Matplotlib.

Publications

Refereed publications

1. Hakobyan, H., et al. (including **Solanki, S.**). *Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. I: Curvilinear Special Relativistic Module.* ApJS, **285**, 11 (2026). arXiv:2511.17710.
2. **Solanki, S.**, Davelaar, J., Ripperda, B., Philippov, A. *Modeling of Lightcurves from Reconnection-powered Very High-energy Flares from M87*.* ApJ, **985**, 147 (2025). arXiv:2504.16298.
3. **Solanki, S.**, et al. *Short-period Heartbeat Binaries from TESS Full-frame Images.* ApJS, **276**, 17 (2025). arXiv:2407.14421.
4. Olmschenk, G., et al. (including **Solanki, S.**). *Short-Period Variables in TESS Full-Frame Image Light Curves Identified via Convolutional Neural Networks.* AJ, **168**, 83 (2024). arXiv:2402.12369.
5. **Solanki, S.**, Ressler, S. M., Murchikova, L., Stone, J. M., Morris, M. R. *The Inner 2 pc of Sagittarius A*: Simulations of the Circumnuclear Disk and Multiphase Gas Accretion in the Galactic Center.* ApJ, **953**, 22 (2023). arXiv:2301.07735.
6. **Solanki, S.**, et al. *Periodicities in the K2 light curve of HP Librae.* MNRAS, **500**, 1222–1230 (2021). arXiv:2010.09754.

Preprints and submissions

1. **Solanki, S.**, Mahlmann, J., Philippov, A. *Nonlinear Decay of Fast Magnetosonic Waves through Weak Turbulence: Force-Free Electrodynamics Simulations.* Submitted to ApJ (2026). arXiv:2606.19450.
2. **Solanki, S.**, Mahlmann, J., Philippov, A., Beloborodov, A. *Damping of Fast Radio Bursts in the Inner Magnetospheres of Magnetars.* Submitted to ApJ Letters (2026). arXiv:2606.19448.
3. Skrabacz, E., Murchikova, L., Ressler, S. M., Ukani, A., **Solanki, S.** *Stellar-wind Fueled Accretion onto Sagittarius A* in the Presence of a Nuclear Star Cluster.* 2026. arXiv:2601.09799.

Awards and honors

- **Burgers Symposium Winner** 2025
  Best poster in the College of Mathematical and Natural Sciences.
- **John Chi-Lin Wang Award** 2024
  Astronomy Department, University of Maryland.
- **Academic High Honors Award** 2021
  Physics Department, University of California, Santa Barbara.
- **Research High Honors Award** 2021
  Physics Department, University of California, Santa Barbara.

Leadership, teaching, and outreach

- **DC Tutoring and Mentoring Initiative** 2025 – Present
  Mentor in the Listo Program.
- **Graduate Student Council** 2023 – 2024
  Astronomy Department, University of Maryland.
- **Cambridge Center for International Research** 2023
  Teaching Assistant.
- **Campus Learning and Assistance Services** 2019 – 2021
  Tutor, University of California, Santa Barbara.

Invited talks and research visits

- **Kavli Institute for Theoretical Physics, UC Santa Barbara** 2025
  Invited Graduate Student Visitor.
- **Princeton Plasma Theory Group, Princeton University** 2024
  Invited Seminar Speaker.
- **Center for Computational Astrophysics, Flatiron Institute** 2024
  SPA Seminar Speaker.
- **Institute for Advanced Study, Princeton** 2024
  Coffee Talk Speaker.

Summer schools and workshops

- **SHIELD Plasma Summer School, Boston** 2026
  Participant.
- **PLASMA Science Summer School, Porquerolles, France** 2024
  Participant.
- **Les Houches Summer School in Plasma Physics, France** 2023
  Participant and Poster Presenter.
- **International High Performance Computing Summer School** 2023
  Participant and Poster Presenter.

In the news

Monthly Roundup: The Supermassive Black Hole of Messier 87. AAS Nova, June 2025.